

**For the submission of your work:**

- Create a folder named **RollNo\_Name\_DBI202\_PaperNo**, e.g. se01245\_LongNT\_DBI202\_01. **Do not** create any subfolder in this folder. All file created will be located in the above folder.

- For each question, you are required to write a database script. Create a file with the name corresponding to the index of the question. For example, **for question 1**, we will create a file named **Q1.sql** and create a file **Q2.sql for question 2**. So, if you do 10 questions, your folder must contain **only** 10 files Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql, Q9.sql and Q10.sql.

- Do not use any commands having the database name such as create database, alter database, use [database name], *etc.*

- Your response must contain only necessary commands for answering the question. **Do not include any other command( GO, EXEC, USE,...)**. For example, if you are required to create a trigger/procedure, then your response should contain only commands for creating the corresponding trigger/procedure; all commands for testing the created trigger/procedure are forbidden.

- On completion, import your work by browsing to the above folder.

**- Note that:**

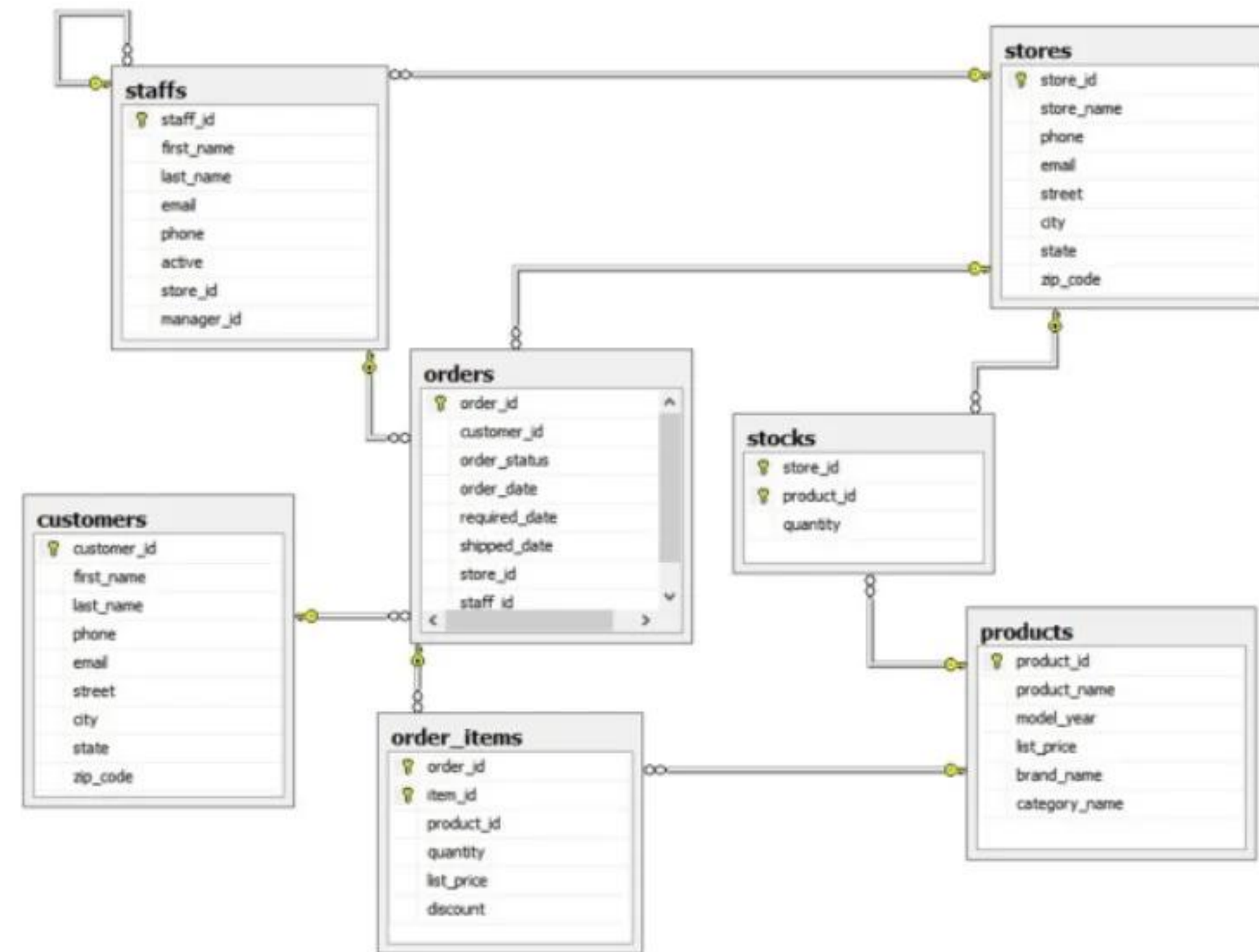
**+ You could use only SQL Server, SQL Server Management Studio, window explorer, winzip.**

**+ Only use all build-in functions of the sql server 2008 for writing queries.**

**+ If any of the previous requirements is not respected, your mark will be 0.**

**From the 2nd question**, you should use the database provided in the .sql file which has the following database diagram. Please, run the provided script to create tables and insert data into your database.







**Question 1:**

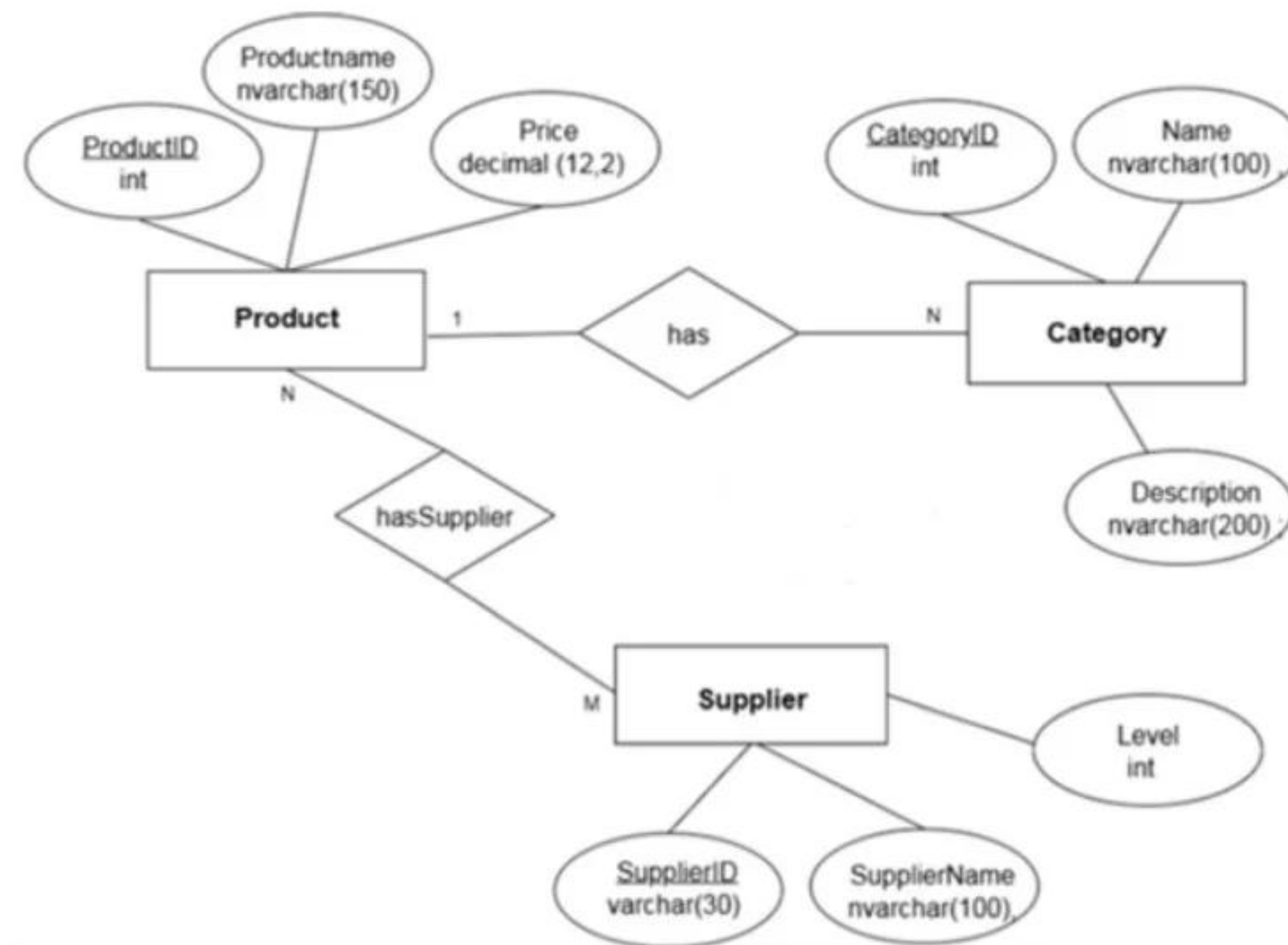
[1.5 marks] Create one database and then write SQL statements to create, in this database, all tables derived from the ERD given in the following picture with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as request, you MUST keep the name of tables, relationship, attributes and data type of attributes as SAME as given in the given ERD.

Attributes with underline belong to the Primary Key of each entity.

Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database\_name" or any statements using database's name in your submission.



Picture 1.1

**Question 2:**



Picture 1.1

**Question 2:**

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[0.5 mark] Write a query to list Customers living in New York . The result includes columns: customer\_id, first\_name, last\_name, email and sort in ascending order by last\_name, first\_name as follows.

|   | customer_id | first_name | last_name | email                     |
|---|-------------|------------|-----------|---------------------------|
| 1 | 1           | Liam       | Johnson   | liam.johnson@cust.example |
| 2 | 2           | Emma       | Smith     | emma.smith@cust.example   |

Picture 2.1

**Question 3:**

[1 mark]. Write a query to list products with a product\_name containing the word ' City'. the result includes: product\_id, product\_name, and list\_price columns, and sort in ascending order by product\_name.as follows.

|   | product_id | product_name | list_price |
|---|------------|--------------|------------|
| 1 | 6          | City Lite    | 259.00     |
| 2 | 7          | City Pro     | 399.00     |
| 3 | 11         | E-City 250   | 1299.00    |

Picture 3.1



**Question 3:**

[1 mark]. Write a query to list products with a product\_name containing the word ' City'. the result includes: product\_id, product\_name, and list\_price columns, and sort in ascending order by product\_name.as follows.

|   | product_id | product_name | list_price |
|---|------------|--------------|------------|
| 1 | 6          | City Lite    | 259.00     |
| 2 | 7          | City Pro     | 399.00     |
| 3 | 11         | E-City 250   | 1299.00    |

Picture 3.1

**Question 4:**

[1 mark ]. Write a query to find orders between 2025-08-05 and 2025-08-15 (YYYY-MM-DD). Display the results including the columns order\_id, customer\_id, order\_date, store\_id, staff\_id, and sort in ascending order by the order\_date column as follows.

|   | order_id | customer_id | order_date | store_id | staff_id |
|---|----------|-------------|------------|----------|----------|
| 1 | 5        | 5           | 2025-08-05 | 5        | 5        |
| 2 | 6        | 6           | 2025-08-06 | 6        | 6        |
| 3 | 7        | 7           | 2025-08-07 | 7        | 7        |
| 4 | 8        | 8           | 2025-08-08 | 8        | 8        |
| 5 | 9        | 9           | 2025-08-09 | 9        | 9        |
| 6 | 10       | 10          | 2025-08-10 | 10       | 10       |
| 7 | 11       | 11          | 2025-08-11 | 11       | 11       |



|    | order_id | customer_id | order_date | store_id | staff_id |
|----|----------|-------------|------------|----------|----------|
| 1  | 5        | 5           | 2025-08-05 | 5        | 5        |
| 2  | 6        | 6           | 2025-08-06 | 6        | 6        |
| 3  | 7        | 7           | 2025-08-07 | 7        | 7        |
| 4  | 8        | 8           | 2025-08-08 | 8        | 8        |
| 5  | 9        | 9           | 2025-08-09 | 9        | 9        |
| 6  | 10       | 10          | 2025-08-10 | 10       | 10       |
| 7  | 11       | 11          | 2025-08-11 | 11       | 11       |
| 8  | 12       | 12          | 2025-08-12 | 12       | 12       |
| 9  | 13       | 13          | 2025-08-13 | 13       | 13       |
| 10 | 14       | 14          | 2025-08-14 | 14       | 14       |
| 11 | 15       | 15          | 2025-08-15 | 15       | 15       |

Picture 4.1

**Question 5:**

[1 mark] Write a query to find the products with the lowest list\_price. Display the results with the columns product\_id, product\_name, and list\_price as follows.

|   | product_id | product_name | list_price |
|---|------------|--------------|------------|
| 1 | 10         | Kids Fun 14  | 149.00     |
| 2 | 13         | BMX Spark    | 149.00     |

Picture 5.1

**Question 6:**

[1 mark]. Write a query to calculate the total quantity in each store. The result includes columns such as store\_id, store\_name, and total\_qty\_in\_stock, then sort in descending order by the total\_qty\_in\_stock column and ascending order by the store\_id column as follows.

|   | store_id | store_name        | total_qty_in_stock |
|---|----------|-------------------|--------------------|
| 1 | 1        | Downtown Cycles   | 17                 |
| 2 | 2        | Riverside Wheels  | 16                 |
| 3 | 4        | Desert Trail      | 13                 |
| 4 | 7        | Seaside Spokes    | 13                 |
| 5 | 3        | Bay City Bikes    | 11                 |
| 6 | 6        | Music City Pedals | 11                 |
| 7 | 5        | Capitol Ride      | 9                  |



**Question 6:**

[1 mark]. Write a query to calculate the total quantity in each store. The result includes columns such as store\_id, store\_name, and total\_qty\_in\_stock, then sort in descending order by the total\_qty\_in\_stock column and ascending order by the store\_id column as follows.

|   | store_id | store_name        | total_qty_in_stock |
|---|----------|-------------------|--------------------|
| 1 | 1        | Downtown Cycles   | 17                 |
| 2 | 2        | Riverside Wheels  | 16                 |
| 3 | 4        | Desert Trail      | 13                 |
| 4 | 7        | Seaside Spokes    | 13                 |
| 5 | 3        | Bay City Bikes    | 11                 |
| 6 | 6        | Music City Pedals | 11                 |
| 7 | 5        | Capitol Ride      | 9                  |

Picture 6.1

**Question 7:**

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[1 mark] . Write a query to find the highest list\_price of products within each category\_name. The result includes columns: category\_name, product\_id, product\_name, and list\_price, then sort in ascending order by category\_name and product\_id as follows.

|    | category_name | product_id | product_name   | list_price |
|----|---------------|------------|----------------|------------|
| 1  | BMX           | 13         | BMX Spark      | 149.00     |
| 2  | City          | 7          | City Pro       | 399.00     |
| 3  | E-Bike        | 12         | E-Mountain 500 | 2199.00    |
| 4  | Fixie         | 15         | Fixie Basic    | 219.00     |
| 5  | Folding       | 9          | Folding Go     | 379.00     |
| 6  | Gravel        | 8          | Gravel GX      | 999.00     |
| 7  | Hybrid        | 17         | Hybrid H3      | 649.00     |
| 8  | Kids          | 10         | Kids Fun 14    | 149.00     |
| 9  | Mountain      | 5          | Mountain X2    | 799.00     |
| 10 | Road          | 3          | Roadster 300   | 699.00     |
| 11 | Touring       | 14         | Touring T1     | 899.00     |



[1 mark] . Write a query to find the highest list\_price of products within each category\_name. The result includes columns: category\_name, product\_id, product\_name, and list\_price, then sort in ascending order by category\_name and product\_id as follows.

|    | category_name | product_id | product_name   | list_price |
|----|---------------|------------|----------------|------------|
| 1  | BMX           | 13         | BMX Spark      | 149.00     |
| 2  | City          | 7          | City Pro       | 399.00     |
| 3  | E-Bike        | 12         | E-Mountain 500 | 2199.00    |
| 4  | Fixie         | 15         | Fixie Basic    | 219.00     |
| 5  | Folding       | 9          | Folding Go     | 379.00     |
| 6  | Gravel        | 8          | Gravel GX      | 999.00     |
| 7  | Hybrid        | 17         | Hybrid H3      | 649.00     |
| 8  | Kids          | 10         | Kids Fun 14    | 149.00     |
| 9  | Mountain      | 5          | Mountain X2    | 799.00     |
| 10 | Road          | 3          | Roadster 300   | 699.00     |
| 11 | Touring       | 14         | Touring T1     | 899.00     |

Picture 7.1

**Question 8:**

[ 1 mark] .Create a store procedure named proc\_SumQuantityProduct to calculate the total products quantity of each store\_id.

Where @Store\_id int is an input parametter and @SumQuantity Decimal(10,2) is an output parameter of the procedure.

For example, when we execute the procedure proc\_SumQuantityProduct by using the following statements the result should be as in the following figure.

```
DECLARE @x DECIMAL(10,2)
```

```
EXEC proc_SumQuantityProduct 2, @x OUTPUT
```

```
SELECT @x SumQuantityProduct
```



| SumQuantityProduct |       |
|--------------------|-------|
| 1                  | 16.00 |

Picture 8.1

**Question 9:**

[1 mark] . Write a query to list the staffs who processed orders for customers in the state of 'CA'

|   | staff_id | first_name | last_name |
|---|----------|------------|-----------|
| 1 | 3        | Chi        | Le        |
| 2 | 11       | Minh       | Dang      |

Picture 9.1

**Question 10:**

[ 1 mark] . Write a query to delete rows from the stocks table whose product\_id belongs to category\_name is 'Mountain'